ISM Note 3

Yang Chen

3/12/2026

1. ISM observables (continued)

Optical

For optical, we can observe the $10^3 - 10^4$ K gas. We can plot the collisional ionization equilibrium curve, which is as follows

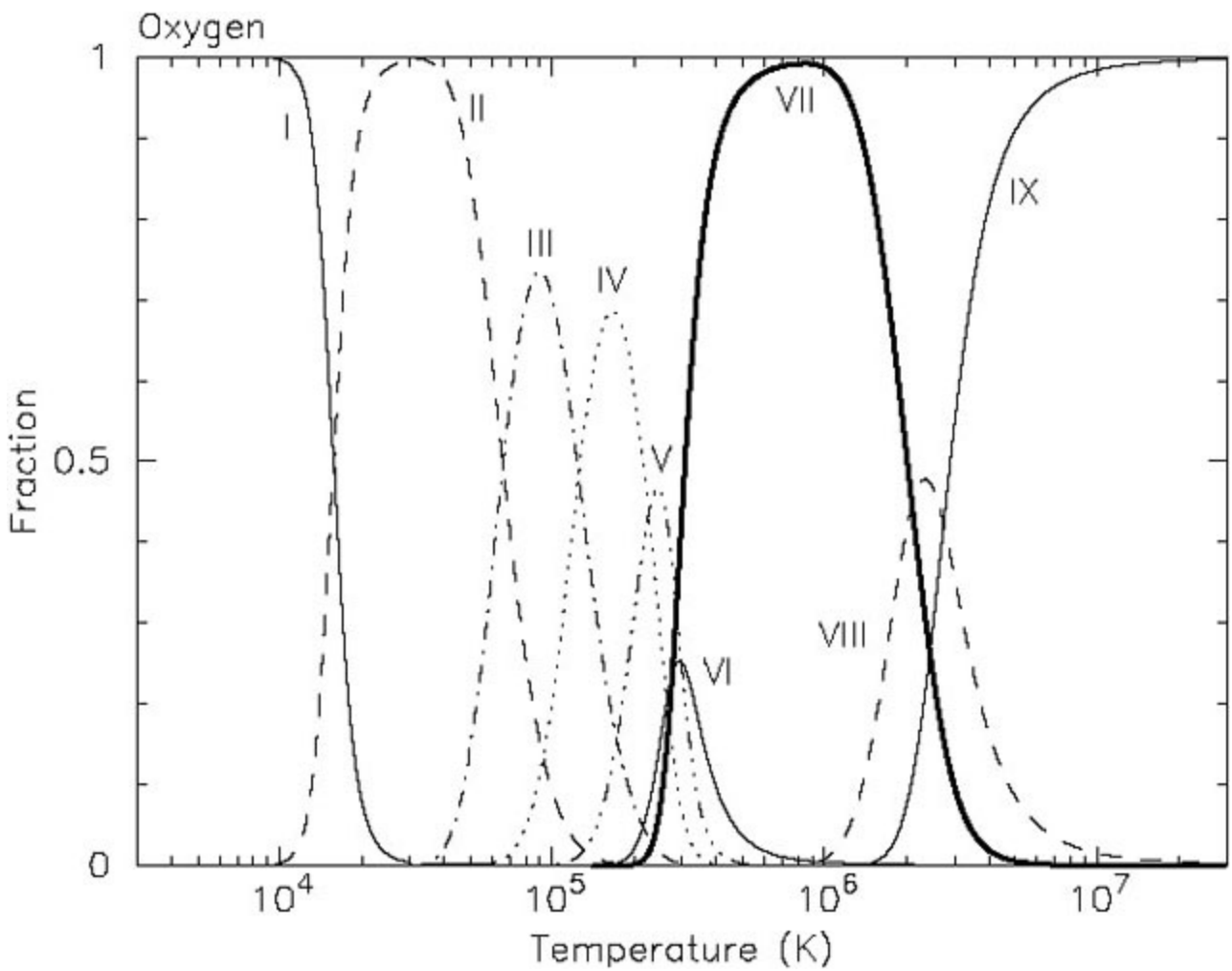

Fig 1. The collisional ionization equilibrium curve.

For ionization, except the collisional ionization, there is also another ionization process called **photoionization**. As the name suggests, it is caused by the photons. This process not only depends on the temperature but also depends on the density of the medium.

Usually in the dilute ISM, we have the relation $t_{\mathrm{ion}} < t_{\mathrm{rec}}$, which means the gas is so dilute that it is hard for the electrons to find ions to recombine. For equilibrium, we assume $t_{\mathrm{rec}} \ll t_{\mathrm{dyn}}$, which means the recombination time scale is much smaller than the dynamical time scale. Physically, this means no matter how fast you drive the gas, the electrons can always find ions to recombine to reach the equilibrium. This is a necessary condition for us to assume the ionization equilibrium.

To give a more intuitive understanding, let's consider the relation $t_{\mathrm{ion}} < t_{\mathrm{rec}}$ in another way. Physically, this implies that an atom gets ionized faster than an ion can find an electron to recombine with. This situation leads to a high ionization fraction, meaning **the gas is predominantly ionized**. This is common in the dilute ISM where the low density makes recombination a slow process (i.e., a long t_{rec}).

Moreover, the relation $t_{\mathrm{rec}} \ll t_{\mathrm{dyn}}$ signifies that the ionization state of the gas can reach equilibrium (balancing ionization and recombination) much faster than the gas can move or be rearranged by large-scale forces (like gravity or turbulence). **This allows us to assume ionization equilibrium when studying the gas**, as the atomic processes are effectively instantaneous compared to the gas dynamics.

Radio

For radio, we usually mean HI ($T \leq 8,000K$), the famous example for this is the 21 cm line. The 21 cm line is a ground state hyperfine structure transition, and it is in the form of both emission and absorption. The abosorption process can be understand in a way shown in the below figure

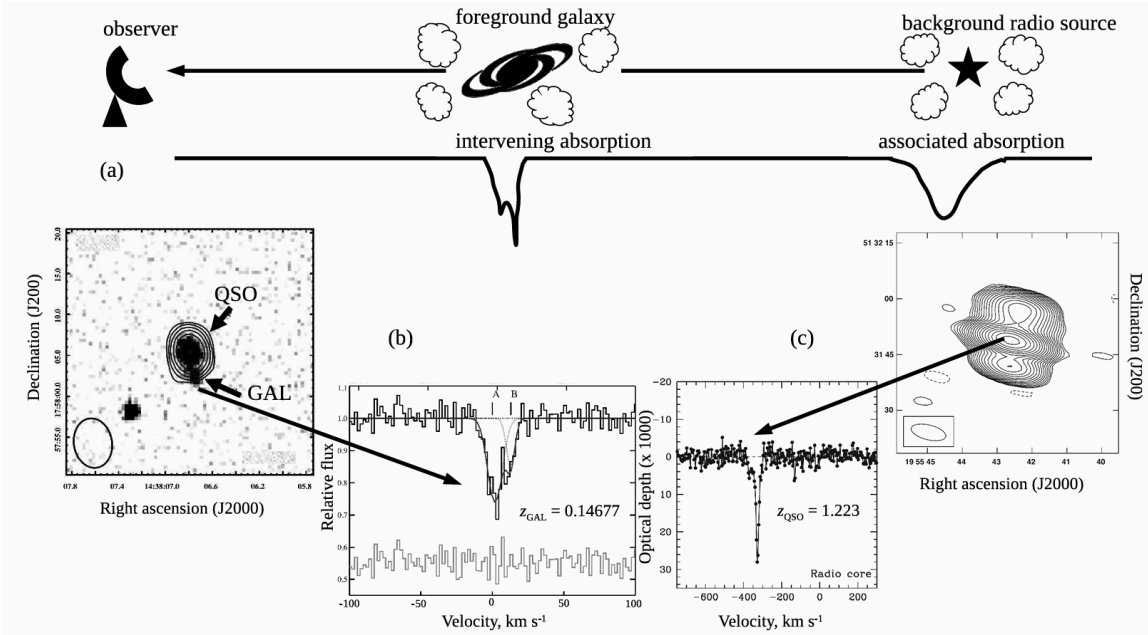

Fig 2. The 21 cm line.

The spectrum shows many absorption lines (not singly the 21 cm line) because of the red shift. And the above figure also shows the 21 cm line from the Quasar/AGN radio source and the intervening galaxy, which is a very good example for the absorption process.

Another important radio emission process is the synchrotron emission, which is caused by the relativistic electrons spiraling around the magnetic field lines. If we drive the electron in a higher energy, then obviously the emission will be in a higher frequency as well, coming over the radio band and into the X-ray band.

Infrared

For infrared band, we can observe the $10 - 10^2$ K dust grains. They are solids embedded in the ISM.

Millimeter

For millimeter band, what we can observe is usually the emission caused by the molecular gas H_2 rotational transition. The molecular gas is important for star formation, so the millimeter band is a good probe for the star formation.

UV

For UV band, we can observe the $10^5 - 10^6$ K gas.

X-ray & Gamma-ray

For X-ray, we can observe the $\geq 10^6$ K ICM gas. So the question is, why can we see the X-ray easier in the ICM than in the ISM? Physically speaking ,the reason is that the largest gravity bounded structure in the universe is the galaxy cluster, and the gas around the galaxy cluster must be in very high temperature to against the gravity, so the gas in the galaxy cluster is usually hotter than that inside the galaxy.

There is another source for the X-ray band, which is SNR (supernova remnants). The SNR is the structure left behind after a star explodes in a supernova. The explosion ejects the star's material into the surrounding interstellar medium at extremely high speeds, creating a powerful shockwave. This expanding shockwave sweeps up interstellar gas and heating it to temperatures of millions of degrees, which can then emit X-rays.

One more source for the X-ray band is the cosmic rays, which are constituted of highly relativistic particles. Cosimic rays are not "rays" in the sense of light, but are overwhelmingly composed of atomic nuclei(about 90% are protons, 9% are helium nuclei and the rast are heavier elements), along with electrons, positrons, and other subatomic particles. The cosmic rays can heat up the ISM and then produce X-ray emission.

2. Observe the ISM indirectly

Plasma propagation (radio signals)

Plasma propagation is the process of radio signals propagating through the plasma. A plasma is a mix of free electrons and ions, thus it can interact with the electromagnetic waves and cause some changes to the EM waves. Qualitatively speaking, this interaction has two main consequences for the wave: one is dispersion, which means the phase

velocity of the wave changes and smears out due to the frequency-sensitive response of the electrons in the plasma. The other is Faraday rotation, which is due to the different phase velocities for left and right circularly polarized waves in the presence of a magnetic field, causing the plane of polarization to rotate as the wave travels through the plasma.

To make the concept of plasma propagation clear and quantitative, astronomers introduce the **dispersion measure**, defined by $DM \propto \int n_e dl$, which is an integral along the line of sight. Here n_e is the electron number density. As we can see, the dispersion measure is a direct measure of the total amount of intervening plasma, and its primary effect is the frequency-dependent time delay of a signal.

Another measure is the **rotation measure**, which is defined by $RM \propto \int n_e B_{\|} dl$, where $B_{\|}$ is the magnetic field parallel to the line of sight. The rotation measure quantifies the amount of Faraday rotation which describes the change of the polarization angle of radio. This rotation is the so-called Faraday rotation.

If we divide the rotation measure by the dispersion measure, we can get an estimate of the parallel magnetic field, which is given by $\langle B_{\|} \rangle = \frac{RM}{DM}$. If $\langle B_{\|} \rangle$ is small, this might not mean the magnetic field strength is small, it might mean the magnetic field is quite tangled and random, so the positive and negative contributions could cancel out each other.

One more measure is the **emission measure**, which is defined by $EM \propto \int n_e^2 dl$. The emission measure is a measure of the intensity of thermal emission from the plasma itself, particularly through a process called thermal bremsstrahlung("free-free emission"). EM tells us about the density and clumpiness of the plasma. Comparing EM and DM can reveal whether the plasma is smoothly distributed or concentrated in clumps.

3. Physical properties of the ISM

The multiphase ISM

Many years after completing this course, you might forget most of the content of this course, but the only thing you need to remember is the concept of the multiphase ISM. The

multiphase here means that the ISM consists of multiple distinct phases of matter, each with different physical properties.

One phase is **neutral gas** (atoms), e.g. HI, whose temperature is about $10^3 - 10^4$ K. One is **molecular gas**, e.g. H_2, whose temperature is about $10 - 10^2$ K. One is **warm ionized gas**, namely WIM, e.g. HII, whose temperature is about $10^4 - 10^5$ K. One is **hot ionized gas**, whose temperature is about $\geq 10^6$ K. The multiphase gas should maintain the thermal pressure equilibrium, which means the number density of different phases should be different, and simply satisfying $n_1 T_1 = n_2 T_2$, where n is the number density and T is the temperature.

The morphology of H_2 is clouds, filaments and sheets. This is because molecular hydrogen can only form and survive in the coldest ($10 - 20$ K) and densest regions of the ISM, where gas and dust can shield the fragile H_2 molecules from destructive ultraviolet radiation. The interplay of galactic turbulence and gravity naturally sculpts this cold gas. Large-scale shockwaves compress the gas into vast sheets, which then collapse under their own gravity to form a network of interconnected filaments. These filaments then fragment into the dense clumps and cores that we call molecular clouds, which are the birthplaces of stars. Therefore, the shape of molecular gas structures directly traces the underlying physical processes of compression and gravitational collapse in the ISM.

One figure for the multiphase ISM is as follows

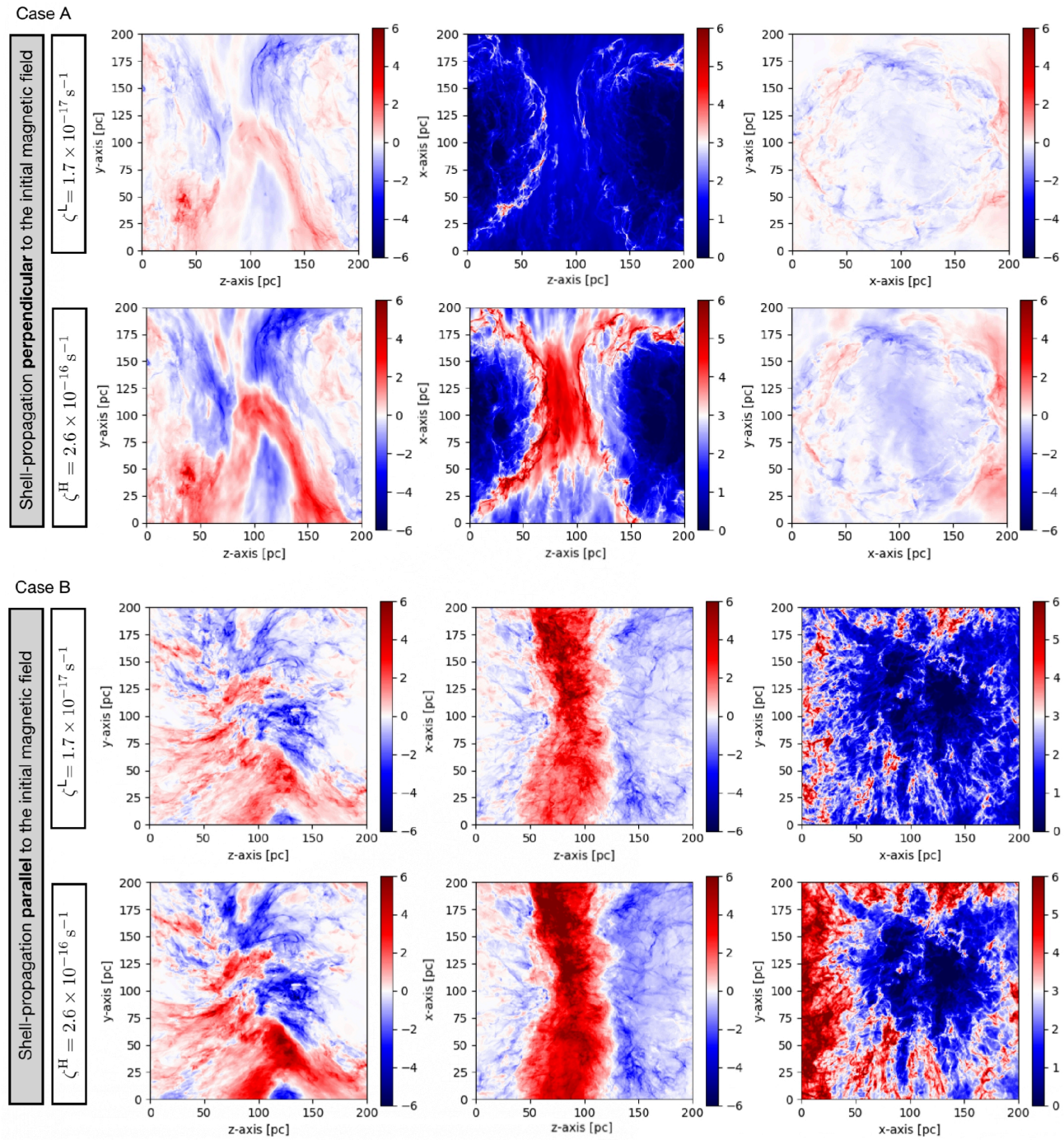

Fig 3. The multiphase ISM.

Other components of the ISM

The ISM also contains the **dust grains**, and the ratio of the dust mass to the gas mass is about 1%. Though the dust grains are in such a small fraction, it can play a significant role in the ISM. The dust grains can absorb or scatter stellar light, and radiate in the infrared, this is called **dust obscured**. The dust contain the vast majority of heavy elements (metals), which can provide cooling for the ISM. This phenomenon is known as **depletion**.

In the cooler and denser phases of the ISM, atoms of elements like carbon, silicon, magnesium, and iron collide with dust grains and stick to their surfaces, effectively removing these atoms from the gas phase.

The ISM also contains the **cosmic rays**, which are highly relativistic particles as we have introduced above. The energy range (per particle) of the cosmic rays is about $10^9 - 10^{21}$ eV, which is a wide range. Thus we can divide the cosmic rays into many categories according to their energy. The first type is the low-energy cosmic rays, which are about $10^9 - 10^{11}$ eV ($1 - 100Gev$), the second type is the high-energy cosmic rays, which are about $10^{11} - 10^{15}$ eV ($100Gev - 1PeV$), and the third type is the ultra-high-energy cosmic rays, which are about $\geq 10^{15}$ eV (1 PeV and above). We can draw the energy spectrum of the cosmic rays, which is as follows

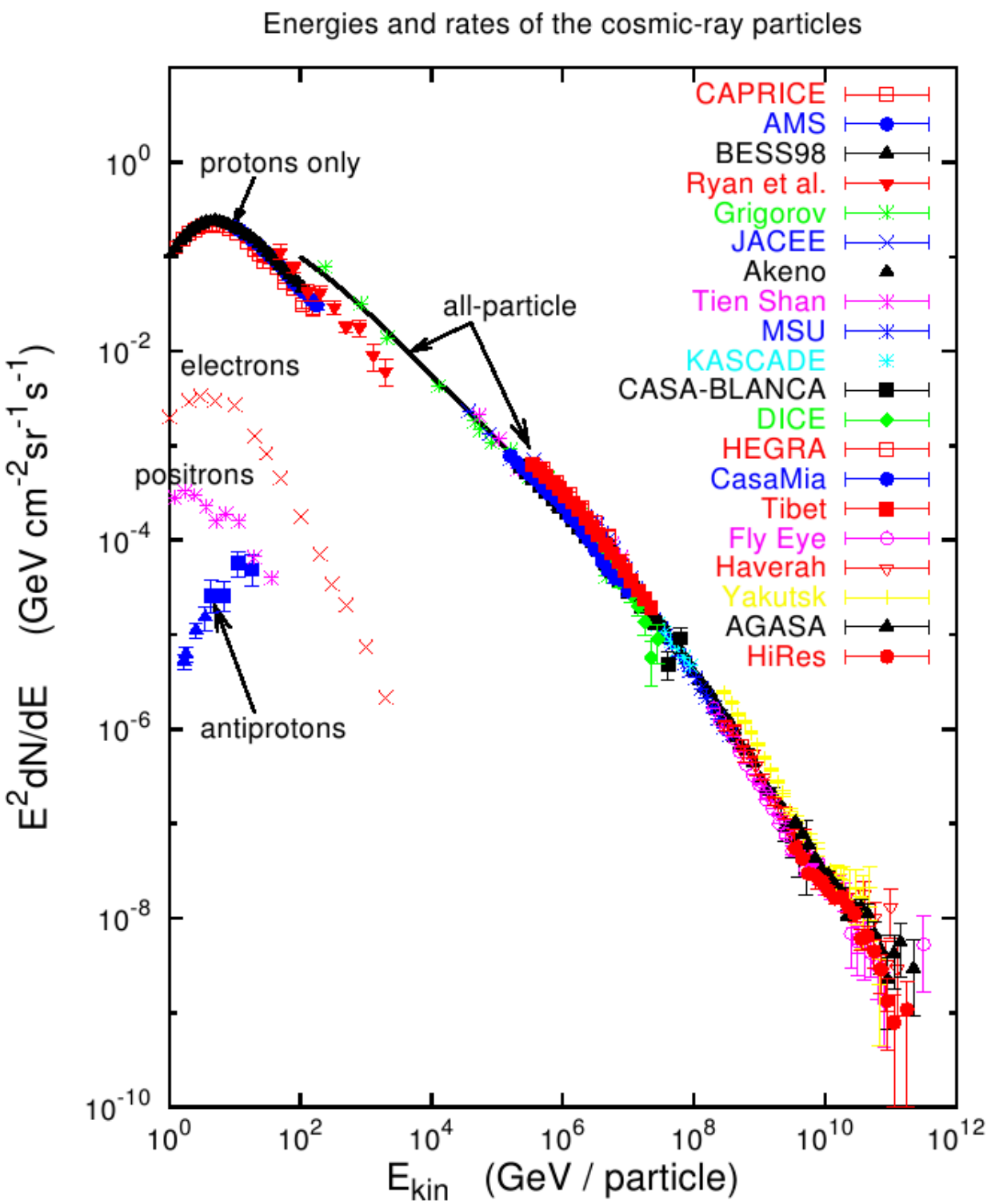

Fig 4. The energy spectrum of the cosmic rays.

From the figure, it is not hard for us to understand that in ISM we usually care about the ~ Gev cosmic rays, which are in higher population. The cosmic ray energy density is about $1eV/cm^3$.

Magnetic field in the ISM

The magnetic field in the ISM is about a few μG. The energy density of the magnetic field is also about $1eV/cm^3$, which is comparable to the energy density of the cosmic rays and the thermal energy density of the ISM. This is called **the equipartition of energy**. In the ISM, the energy equipartition is between the kinetic energy, the thermal energy, and the non-thermal energy (cosmic rays and magnetic field).

Stuff related to stars

HII regions are the ionized gas around the young stars. We know the star formation happens in the molecular gas, but once the stars are formed, they can emit lights and ionize the gas around them, and then form the HII regions. The HII region is also called **the Strömgren sphere**, which is a sphere of ionized gas around the star. The radius can be calculated analytically, but we will leave it for the future course.

Another important thing is **the planetary nebular**, which is the gas ejected by the dying stars. One more important thing is the supernova remnants (SNR), which is the gas ejected by the supernova explosion as we discussed above. We can observe the SNR in different bands, such as radio, optical and X-ray, because the shock wave can heat the gas very efficiently and emit various wavelength radiations. The SNR is also an important source for the cosmic rays. One supernova can release about 10^{51} erg energy, and only about **10% - 30%** of the energy can be converted into the cosmic ray energy, other energy will be converted into the thermal energy and the kinetic energy of the ISM.

One interesting thing is that the cosimic rays energy fraction is only about 10% of the supernova energy, which is much smaller than the thermal energy fraction, this seems to contradict with the equipartition of energy.

The key point here is that the cosmic rays have a much longer lifetime than the thermal energy. The thermal energy in the ISM is mainly from the supernova explosions, which can heat up the gas to very high temperatures. However, this thermal energy is lost relatively quickly through radiative cooling processes, as hot gas emits photons and cools down on timescales of millions of years or less, depending on the density and temperature. In

contrast, cosmic rays are highly relativistic particles that can travel vast distances and remain confined within the galaxy by magnetic fields for tens to hundreds of millions of years. Because cosmic rays lose their energy much more slowly (mainly through interactions with the ISM and escape from the galaxy), their energy accumulates over time.

Molecular clouds are mainly composed of H_2, and they are the birthplace of stars. There are also some other molecules in the molecular clouds, such as CO, which is a good tracer for the molecular gas. The temperature of the molecular clouds is about $30 - 10^2$ K,and the typical mass is about $10^5 M_\odot$. Sometimes it can form a big molecular cloud, which is called the **giant molecular cloud** (GMC). The GMC is the collapse beds to star formation, and the typical mass of the GMC is about $10^5 - 10^6 M_\odot$.

We have talked about many things about star formation regions, like the molecular clouds mentioned above. Then one question is: what is the efficiency of star formation? Typically speaking, it is a few percent, which means only a few percent of the gas in the molecular clouds can be converted into stars. Then one straightforward question is: why is the star formation efficiency so low? In other words, what can prevent the gas from collapsing and form the stars? There are many factors that can play an important role: rotations, magnetic fields, turbulence, etc. But we will not talk about these factors in detail, and we will leave it for the future course.

4. ISM ecology

It is important to point out that ISM is not static but highly dynamic and turbulent. The ISM can form stars and the stars can form stellar winds from the live stars and ejecta (supernova) from the dying stars. The feedback process from the ejecta is called metal enrichment, which can enrich the ISM with metals. One sketch for the ISM ecology is as follows

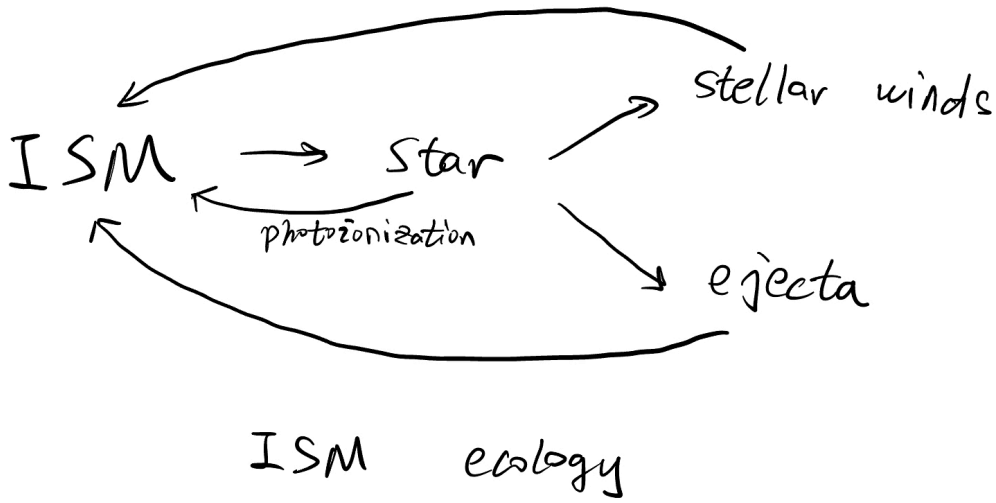

Fig 5. The ISM ecology.

ISM in ellipticals

In the previous course, we know that the elliptical galaxies have low SFR. (This course ends here, and the content is continued in the next course)